

A Trace-Ideal Completion of the Intrinsic Bulk Two-Step Operator

Hilbert–Schmidt Convergence, Trace-Norm Squares, and Fredholm Determinants

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Abstract

An intrinsic rank-two subtraction has isolated the Perron and negative parity modes of a folded-Gaussian Markov operator at fixed noise $\sigma = 10^{-2}$. The resulting bulk operator

$$\mathcal{B} = \mathcal{K} - \mathcal{Q}_+ - \mathcal{Q}_-$$

is the natural one-step complement, but convergence of ordinary Fredholm determinants would require trace-norm control that is not supplied by the piecewise-constant discretization. We close the trace-ideal gap by passing to the intrinsic two-step operator.

For the full exact-real midpoint family and the adaptive Gaussian-cutoff family, lifted to $L^2([0, 1])$, we prove

$$\|B_n - \mathcal{B}\|_{\mathfrak{S}_2} = O(n^{-1}).$$

Since products of two Hilbert–Schmidt operators are trace class,

$$\|B_n^2 - \mathcal{B}^2\|_{\mathfrak{S}_1} \leq \|B_n - \mathcal{B}\|_{\mathfrak{S}_2} (\|B_n\|_{\mathfrak{S}_2} + \|\mathcal{B}\|_{\mathfrak{S}_2}) = O(n^{-1}).$$

Consequently every fixed even bulk trace converges, and the Fredholm determinants

$$\mathcal{D}_{n,\text{bulk}}(w) = \det(\mathbf{I} - wB_n^2)$$

converge locally uniformly on the entire w -plane to $\mathcal{D}_{\text{bulk}}(w) = \det(\mathbf{I} - w\mathcal{B}^2)$. The construction is exactly the symmetric completion of the regularized one-step determinant:

$$\det_2(\mathbf{I} - z\mathcal{B}) \det_2(\mathbf{I} + z\mathcal{B}) = \det(\mathbf{I} - z^2\mathcal{B}^2).$$

The validated continuum bound is $\|\mathcal{B}\|_{\mathfrak{S}_2} \leq 15.418003788081082$. At $n = 65536$ the outward bounds are 0.1281382922864061 in Hilbert–Schmidt norm and 3.967692773690177 in square trace norm. At $n = 2^{30}$ they decrease to $3.785070933298939 \times 10^{-6}$ and $1.167164903022793 \times 10^{-4}$; on $|w| \leq 10^{-2}$ the corresponding uniform determinant error is at most $3.682918268863141 \times 10^{-4}$. A separate floating sparse-LU pilot at 2048, 4096, and 8192 shows a mean dyadic determinant-difference ratio 0.2493, but this sharper center behavior is not used in the theorem.

All results are at fixed positive noise. No zero-noise limit, arithmetic trace formula, prime-power identity, zeta-zero correspondence, self-adjoint Hilbert–Pólya operator, or Riemann-hypothesis conclusion is asserted.

Keywords: Hilbert–Schmidt operator; trace ideal; Fredholm determinant; regularized determinant; Riesz projection; Markov operator; Nyström approximation.

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1 Introduction

The useful spectral object in a noisy dynamical system is often not the full transfer operator but the complement left after isolated peripheral modes have been removed. For the folded-Gaussian

Markov operator studied in this series, the Perron eigenvalue 1 and a simple negative resonance near -1 have now been isolated by explicit Euclidean contours. Their weighted Riesz terms are intrinsic, real, smooth, and gauge-free. Thus the one-step bulk operator

$$\mathcal{B} = \mathcal{K} - \mathcal{Q}_+ - \mathcal{Q}_- \quad (1)$$

is no longer a numerical ansatz: it is a rigorously defined continuum operator with validated full and sparse discretizations [4, 5].

That completion leaves a trace-ideal mismatch. The most natural cellwise comparison is Hilbert–Schmidt. If k_n is a piecewise-constant kernel approximation to a smooth kernel k , then

$$\|k_n - k\|_{L^2([0,1]^2)} = O(n^{-1})$$

without requiring pointwise interpolation. Under the standard isometry between matrices and piecewise-constant integral operators, this is exactly Hilbert–Schmidt convergence. It does not by itself imply trace-norm convergence. Therefore one cannot simply pass from $B_n \rightarrow \mathcal{B}$ in $\mathfrak{S}_2$ to local uniform convergence of $\det(\mathbf{I} - zB_n)$.

The two-step operator has the missing regularity automatically. The trace ideal product rule

$$\mathfrak{S}_2 \mathfrak{S}_2 \subset \mathfrak{S}_1$$

turns the elementary factorization

$$B_n^2 - \mathcal{B}^2 = (B_n - \mathcal{B})B_n + \mathcal{B}(B_n - \mathcal{B})$$

into a trace-norm estimate. This is the central mechanism of the paper. It is abstract, short, and independent of normality or self-adjointness. All dynamical work enters through the explicit Hilbert–Schmidt error ledger for $B_n - \mathcal{B}$.

There are three reasons that the squared determinant is the correct next object.

1. It is intrinsic. Both peripheral branches have already been removed by weighted Riesz calculus; no ordering or normalization of eigenvectors remains in the definition.
2. It is trace class under exactly the topology delivered by the discretization. No unproved trace-norm approximation of the one-step operator is inserted.
3. It joins the regularized determinant of the long-cycle analysis by the exact identity

$$\det_2(\mathbf{I} - z\mathcal{B}) \det_2(\mathbf{I} + z\mathcal{B}) = \det(\mathbf{I} - z^2\mathcal{B}^2).$$

The linear regularization terms cancel between z and $-z$.

The distinction between these statements and a Hilbert–Pólya construction is substantial. The operator $\mathcal{B}$ is nonnormal, the determinant is attached to one fixed positive noise width, and no arithmetic trace formula is available. Squaring packages the even spectral traces; it does not make the operator self-adjoint and it does not identify any determinant zero with a Riemann zero. The result established here is a functional-analytic bridge needed before such comparisons can even be formulated cleanly.

Main contribution and evidence levels

The paper keeps three logical layers separate.

Analytic theorem

Hilbert–Schmidt-to-trace-norm squaring, even-trace convergence, Fredholm determinant continuity, and the symmetric $\det_2$ identity.

Outward certificate

Explicit continuum kernel bounds, Hilbert–Galerkin errors, midpoint and row-normalization errors, adaptive Gaussian-cutoff errors, and Perron/parity weighted-Riesz transport for every $n \geq 65536$.

Floating diagnostic

Sparse-LU determinant values for the exact archived binary64 matrices at 2048, 4096, and 8192. These values illustrate stabilization but prove no continuum statement.

2 Operator, contours, and discretizations

Let u_c be the unique root in $(1.5, 1.6)$ of

$$u^3 - 2u^2 + 2u - 2 = 0,$$

fix $\sigma = 1/100$, and put

$$m(x) = 1 - u_c x^2.$$

For $x, y \in [0, 1]$, define

$$g(x, y) = e^{-(y-m(x))^2/(2\sigma^2)} + e^{-(y+m(x))^2/(2\sigma^2)}, \quad (2)$$

$$Z(x) = \int_0^1 g(x, y) dy, \quad k(x, y) = \frac{g(x, y)}{Z(x)}. \quad (3)$$

The folded Markov operator on $H = L^2([0, 1])$ is

$$(\mathcal{K}f)(x) = \int_0^1 k(x, y) f(y) dy. \quad (4)$$

Its kernel is real, strictly positive, and smooth on the compact square, and $\mathcal{K}\mathbf{1} = \mathbf{1}$.

The validated contours are

$$\Gamma_+ = \{z \in \mathbb{C} : |z - 1| = 0.05\}, \quad (5)$$

$$\Gamma_- = \{z \in \mathbb{C} : |z + 0.9865481927458079| = 0.05\}. \quad (6)$$

Each contains one algebraically simple eigenvalue, and the circles are disjoint. Their weighted Riesz terms are

$$\mathcal{Q}_\pm = \frac{1}{2\pi i} \int_{\Gamma_\pm} z(z - \mathcal{K})^{-1} dz. \quad (7)$$

For the Perron branch, $\mathcal{Q}_+$ is also the rank-one projection $f \mapsto \mathbf{1}\langle \pi, f \rangle$, where $\mathcal{K}^*\pi = \pi$ and $\int_0^1 \pi = 1$. For the negative branch, $\mathcal{Q}_- = \lambda_- \mathcal{P}_-$, where $\mathcal{P}_-$ is its Riesz projection. The terms commute with $\mathcal{K}$, annihilate one another, and have smooth real rank-one kernels. We use the bulk definition (1).

2.1 Cell spaces and lifted matrices

Let $I_{n,j} = [j/n, (j+1)/n)$ and let $V_n \subset H$ be the space of functions constant on the n cells. The vectors

$$e_{n,j} = \sqrt{n} \mathbf{1}_{I_{n,j}}, \quad 0 \leq j < n,$$

form an orthonormal basis. Write E_n for the orthogonal projection onto V_n and $J_n : \mathbb{C}^n \rightarrow V_n$ for the corresponding isometry. Every matrix A_n is compared with the continuum through the zero-complement lift

$$\hat{A}_n = J_n A_n J_n^*. \quad (8)$$

Then

$$\|\hat{A}_n\|_{\mathfrak{S}_2} = \|A_n\|_F.$$

We suppress the hat after this convention has been fixed.

Let $x_i = y_i = (i + \frac{1}{2})/n$ and $h = 1/n$. The continuum-normalized midpoint matrix is

$$(P_n^\circ)_{ij} = hk(x_i, y_j). \quad (9)$$

The full exact-real matrix P_n row-normalizes the unnormalized folded Gaussian midpoint weights. The adaptive matrix P_n^{ad} first retains destinations within $L_n\sigma$ of either folded mean and then renormalizes, where

$$L_n = \max\{8, 2\sqrt{\log n}\}. \quad (10)$$

At $n = 2048, 4096, 8192$, one has $L_n = 8$, so the archived fixed-width matrices are also exact members of the declared adaptive sequence.

For $F_n \in \{P_n, P_n^{\text{ad}}\}$, define

$$Q_{\pm, n} = \frac{1}{2\pi i} \int_{\Gamma_{\pm}} z(z - F_n)^{-1} dz, \quad B_n = F_n - Q_{+, n} - Q_{-, n}. \quad (11)$$

The all-grid contour theorem of the preceding work makes these definitions valid for every integer $n \geq 65536$.

3 The abstract trace-ideal mechanism

We write $\mathfrak{S}_2(H)$ and $\mathfrak{S}_1(H)$ for the Hilbert–Schmidt and trace-class ideals. Their norms are denoted by $\|\cdot\|_2$ and $\|\cdot\|_1$ in this section.

Lemma 3.1 (Hilbert–Schmidt square continuity). *If $X, Y \in \mathfrak{S}_2(H)$, then $X^2, Y^2 \in \mathfrak{S}_1(H)$ and*

$$\|X^2 - Y^2\|_1 \leq \|X - Y\|_2 (\|X\|_2 + \|Y\|_2). \quad (12)$$

Proof. Use

$$X^2 - Y^2 = (X - Y)X + Y(X - Y)$$

and the ideal inequality $\|AC\|_1 \leq \|A\|_2 \|C\|_2$ for $A, C \in \mathfrak{S}_2(H)$ [1, 2]. $\square$

Theorem 3.2 (Squares, even traces, and determinants). *Suppose $B_n, B \in \mathfrak{S}_2(H)$ and*

$$\|B_n - B\|_2 \leq \varepsilon_n, \quad \|B\|_2 \leq M. \quad (13)$$

Put

$$\delta_n = \varepsilon_n(2M + \varepsilon_n). \quad (14)$$

Then

$$\|B^2\|_1 \leq M^2, \quad (15)$$

$$\|B_n^2\|_1 \leq (M + \varepsilon_n)^2, \quad (16)$$

$$\|B_n^2 - B^2\|_1 \leq \delta_n. \quad (17)$$

For every fixed integer $m \geq 1$,

$$|\text{tr}(B_n^{2m}) - \text{tr}(B^{2m})| \leq m\delta_n \left[\max\{M^2, (M + \varepsilon_n)^2\} \right]^{m-1}. \quad (18)$$

Moreover, with

$$\mathcal{D}_n(w) = \det(\mathbf{I} - wB_n^2), \quad \mathcal{D}(w) = \det(\mathbf{I} - wB^2), \quad (19)$$

one has, for every $R \geq 0$,

$$\sup_{|w| \leq R} |\mathcal{D}_n(w) - \mathcal{D}(w)| \leq R\delta_n \exp\left(1 + RM^2 + R(M + \varepsilon_n)^2\right). \quad (20)$$

In particular, $\varepsilon_n \rightarrow 0$ implies local uniform convergence $\mathcal{D}_n \rightarrow \mathcal{D}$ on $\mathbb{C}$.

Proof. The first three bounds follow from $\|AC\|_1 \leq \|A\|_2 \|C\|_2$, the triangle inequality, and [lemma 3.1](#). For traces, put $A_n = B_n^2$ and $A = B^2$. The telescoping identity

$$A_n^m - A^m = \sum_{j=0}^{m-1} A_n^j (A_n - A) A^{m-1-j}$$

and $\|RST\|_1 \leq \|R\| \|S\|_1 \|T\|$ give

$$\|A_n^m - A^m\|_1 \leq m \|A_n - A\|_1 \max\{\|A_n\|, \|A\|\}^{m-1}.$$

Use $\|A\| \leq \|A\|_1$ and $|\operatorname{tr} C| \leq \|C\|_1$.

For determinants, the standard trace-ideal continuity estimate is

$$|\det(\mathbf{I} + C) - \det(\mathbf{I} + D)| \leq \|C - D\|_1 e^{1 + \|C\|_1 + \|D\|_1} \quad (21)$$

for trace-class C, D [2]. Apply it to $C = -wB_n^2$ and $D = -wB^2$, then maximize over $|w| \leq R$. $\square$

Remark 3.3 (Why the one-step determinant is not used). At the fixed smooth Gaussian kernel, stronger trace-class regularity of $\mathcal{B}$ is available. What is not supplied by the present discretization ledger is convergence $B_n \rightarrow \mathcal{B}$ in trace norm. The theorem therefore does not infer convergence of $\det(\mathbf{I} - zB_n)$ from Hilbert–Schmidt convergence. Passing to B_n^2 is the rigorously justified trace-ideal completion.

4 Hilbert–Schmidt convergence of the intrinsic bulk

The dynamical part of the argument is to construct an explicit ε_n in (13). It has a Markov-kernel part and two weighted-Riesz parts.

4.1 Continuum to cell-average Galerkin

Let k_n^G be the kernel of $E_n \mathcal{K} E_n$. Applying the cellwise Poincaré–Wirtinger inequality in the source and target variables gives

$$\|\mathcal{K} - E_n \mathcal{K} E_n\|_{\mathfrak{S}_2} = \|k - k_n^G\|_{L^2([0,1]^2)} \leq \frac{1}{\pi n} (\|k_x\|_{L^2} + \|k_y\|_{L^2}). \quad (22)$$

This first-order term is the asymptotic bottleneck of the current proof. It is the natural price of lifting a smooth kernel by cell constants rather than a higher-order basis.

4.2 Midpoint and exact row normalization

The one-dimensional midpoint-average Peano kernel has L^2 norm $h^{3/2}/\sqrt{320}$. Tensor decomposition therefore gives

$$\|E_n \mathcal{K} E_n - P_n^\circ\|_{\mathfrak{S}_2} \leq \frac{h^2}{\sqrt{320}} (\|k_{xx}\|_{L^2} + \|k_{yy}\|_{L^2}) + \frac{h^4}{320} \|k_{xxyy}\|_{L^2}. \quad (23)$$

Let $Z_* > 0$ be the validated normalizer lower bound and put

$$a_n = \frac{e^{-1/2}}{\sigma} h^2, \quad \rho_n = \frac{a_n}{Z_* - a_n}. \quad (24)$$

Exact row normalization is a diagonal left scaling whose relative defect is at most ρ_n . Hence

$$\|P_n - P_n^\circ\|_{\mathfrak{S}_2} \leq \rho_n \left(\|k\|_{L^2} + \|E_n \mathcal{K} E_n - P_n^\circ\|_{\mathfrak{S}_2} \right). \quad (25)$$

Although the inherited certificate field was originally named “spectral norm defect,” its derivation is a relative row-scaling bound times a Frobenius kernel norm. It is therefore also a valid Hilbert–Schmidt upper bound.

4.3 Adaptive Gaussian cutoff

For a declared support multiple L , the omitted row mass obeys

$$\mu_{n,L} \leq \frac{2\sqrt{e} e^{-L^2/2} (h + \sigma/L)}{\sigma - h}. \quad (26)$$

Writing $\alpha_{n,L} = \mu_{n,L}/(1 - \mu_{n,L})$, the omitted and renormalization terms give the Frobenius-square ledger

$$\begin{aligned} \|P_n^{(L)} - P_n\|_{\mathfrak{S}_2}^2 &\leq \frac{4e e^{-L^2} (h + \sigma/(2L))}{(\sigma - h)^2} \\ &\quad + \frac{e\alpha_{n,L}^2 (4h + 2\sqrt{\pi}\sigma)}{(\sigma - h)^2}. \end{aligned} \quad (27)$$

Thus (10) gives

$$\|P_n^{\text{ad}} - P_n\|_{\mathfrak{S}_2} = O\left(n^{-2}(\log n)^{-1/4}\right). \quad (28)$$

The certificate field for (27) is also historically named “spectral norm upper,” but it is explicitly the square root of a Frobenius-square ledger and may be used in $\mathfrak{S}_2$.

4.4 Weighted-Riesz transport

For a circle Γ of radius r , with $R_\Gamma = \max_{z \in \Gamma} |z|$, the resolvent identity gives

$$\|\mathcal{Q}(A; \Gamma) - \mathcal{Q}(C; \Gamma)\| \leq r R_\Gamma M_A M_C \|A - C\|, \quad (29)$$

provided the two contour resolvents are bounded by M_A and M_C . The validated continuum bounds are

$$\sup_{\Gamma_+} \|(z - \mathcal{K})^{-1}\| \leq 81.30575843422578, \quad \sup_{\Gamma_-} \|(z - \mathcal{K})^{-1}\| \leq 112.95503061584434. \quad (30)$$

The comparison between the continuum weighted term and its Galerkin compression uses the orthogonal splitting $H = V_n \oplus V_n^\perp$. The source-to-detail and target-to-detail blocks are $O(n^{-1})$, while the detail block is $O(n^{-2})$. Schur inversion leaves four weighted resolvent blocks. In particular, the top-left Schur inverse is the V_n compression of the complete continuum resolvent, so its norm is at most the corresponding bound in (30). This is the reason the certificate may use the continuum resolvent as the “Schur-inverse upper.”

Let $\eta_{\pm,n}$ denote the complete operator-norm weighted-term error after the complement Schur transport, midpoint transfer, exact normalization, and, when applicable, adaptive cutoff. Since both the continuum and discrete weighted terms have rank one,

$$\text{rank}(Q_{\pm,n} - \mathcal{Q}_\pm) \leq 2, \quad \|Q_{\pm,n} - \mathcal{Q}_\pm\|_{\mathfrak{S}_2} \leq \sqrt{2} \eta_{\pm,n}. \quad (31)$$

All transferred contour gates are closed at $n = 65536$. Every input defect decreases with n , and L_n is nondecreasing, so the same is true for every integer $n \geq 65536$.

Theorem 4.1 (Validated bulk Hilbert–Schmidt convergence). *For the full and adaptive families in (11), and every integer $n \geq 65536$, there are explicit outward numbers $\varepsilon_n^{\text{full}}$ and $\varepsilon_n^{\text{ad}}$ such that*

$$\|B_n^{\text{full}} - \mathcal{B}\|_{\mathfrak{S}_2} \leq \varepsilon_n^{\text{full}} = O(n^{-1}), \quad (32)$$

$$\|B_n^{\text{ad}} - \mathcal{B}\|_{\mathfrak{S}_2} \leq \varepsilon_n^{\text{ad}} = O(n^{-1}). \quad (33)$$

They are composed as

$$\varepsilon_n = \eta_{K,n} + \sqrt{2}(\eta_{+,n} + \eta_{-,n}), \quad (34)$$

where $\eta_{K,n}$ is the appropriate full or adaptive Markov Hilbert–Schmidt defect.

The intrinsic continuum bulk satisfies

$$\|\mathcal{B}\|_{\mathfrak{S}_2} \leq \|\mathcal{K}\|_{\mathfrak{S}_2} + \|\mathcal{Q}_+ + \mathcal{Q}_-\|_{\mathfrak{S}_2} \leq 15.418003788081082. \quad (35)$$

Proof. Add (22), (23), and (25), together with (27) for the adaptive family. Add the two weighted-term errors using (31). The first-order terms are the cellwise Galerkin defect and the continuum-complement weighted transport; all other terms are second order or smaller. The final continuum bound uses the validated values

$$\|\mathcal{K}\|_{\mathfrak{S}_2} \leq 5.498549224049743, \quad \|\mathcal{Q}_+ + \mathcal{Q}_-\|_{\mathfrak{S}_2} \leq 9.919454564031339.$$

□

Remark 4.2 (Fixed eight-sigma support). The fixed-width family remains uniformly spectrally stable and its peripheral terms are well defined. Its omitted Gaussian mass has a nonzero continuum floor, so this paper does not claim that the fixed eight-sigma family converges to $\mathcal{B}$ in Hilbert–Schmidt norm. Adaptive growth is essential to the continuum theorem.

5 Trace-norm squares and even bulk traces

Combining theorems 3.2 and 4.1 immediately closes the two-step trace ideal.

Theorem 5.1 (Validated bulk-square trace-norm convergence). *For the full and adaptive families, every integer $n \geq 65536$ satisfies*

$$\|(B_n^\star)^2 - \mathcal{B}^2\|_{\mathfrak{S}_1} \leq \delta_n^\star := \varepsilon_n^\star (30.836007576162164 + \varepsilon_n^\star), \quad \star \in \{\text{full}, \text{ad}\}. \quad (36)$$

In particular,

$$\|(B_n^\star)^2 - \mathcal{B}^2\|_{\mathfrak{S}_1} = O(n^{-1}). \quad (37)$$

For every fixed $m \geq 1$,

$$\text{tr}((B_n^\star)^{2m}) \longrightarrow \text{tr}(\mathcal{B}^{2m}). \quad (38)$$

The explicit coefficient error is (18) with $M = 15.418003788081082$ and $\varepsilon_n = \varepsilon_n^\star$.

The result is stronger than convergence of a finite list of computed eigenvalues. Trace-norm convergence controls the complete compact spectrum of the two-step operator at the level needed by Fredholm determinants. It also controls each even trace without a spectral diagonalization and without assuming normality.

At the threshold $n = 65536$, the full ledger decomposes as

$$\begin{aligned} \eta_{K,n} &\leq 0.005119125317796473, \\ \eta_{+,n} &\leq 0.026770586570540194, \quad \eta_{-,n} \leq 0.06021710060888367, \\ \sqrt{2}(\eta_{+,n} + \eta_{-,n}) &\leq 0.12301916696860951. \end{aligned} \quad (39)$$

Thus

$$\varepsilon_{65536}^{\text{full}} \leq 0.1281382922864061, \quad \delta_{65536}^{\text{full}} \leq 3.967692773690177.$$

The threshold constants are intentionally conservative: their purpose is to certify the topology and the asymptotic convergence, not to approximate the determinant sharply at the smallest admitted grid.

n	$\varepsilon_n^{\text{full}}$	$\varepsilon_n^{\text{ad}}$	δ_n^{full}	$\Delta_n(10^{-3})$	$\Delta_n(10^{-2})$
2^{16}	1.2814×10^{-1}	1.2814×10^{-1}	3.9677	1.7419×10^{-2}	1.3027×10^1
2^{20}	3.9318×10^{-3}	3.9318×10^{-3}	1.2126×10^{-1}	5.3031×10^{-4}	3.8309×10^{-1}
2^{24}	2.4245×10^{-4}	2.4245×10^{-4}	7.4761×10^{-3}	3.2693×10^{-5}	2.3592×10^{-2}
2^{30}	3.7851×10^{-6}	3.7851×10^{-6}	1.1672×10^{-4}	5.1039×10^{-7}	3.6829×10^{-4}

Table 1: Selected outward ledgers. Here ε_n is the bulk Hilbert–Schmidt error, δ_n is the bulk-square trace-norm error, and $\Delta_n(R)$ is the full-family determinant upper on $|w| \leq R$ from (20). Full and adaptive values agree to the displayed precision because the adaptive cutoff is much smaller than the first-order Galerkin term.

The scaled final values are

$$2^{30} \varepsilon_{2^{30}}^{\text{full}} \leq 4064.188967889787, \quad 2^{30} \delta_{2^{30}}^{\text{full}} \leq 125323.3771880477. \quad (40)$$

They exhibit the certified first-order regime. The constants are larger at 2^{16} because the contour-resolvent transfers have not yet reached their asymptotic values.

6 Fredholm determinants and the symmetric regularization identity

Since $\mathcal{B}^2 \in \mathfrak{S}_1$, define the intrinsic bulk two-step determinant

$$\mathcal{D}_{\text{bulk}}(w) = \det(\mathbf{I} - w\mathcal{B}^2). \quad (41)$$

It is entire in w . In the norm-safe disk $|w| \|\mathcal{B}^2\| < 1$, its logarithm has the trace expansion

$$\log \mathcal{D}_{\text{bulk}}(w) = - \sum_{m \geq 1} \frac{w^m}{m} \text{tr}(\mathcal{B}^{2m}). \quad (42)$$

Thus the coefficients are precisely the even intrinsic bulk traces closed in [theorem 5.1](#).

Theorem 6.1 (Locally uniform determinant convergence). *For $\star \in \{\text{full}, \text{ad}\}$, put*

$$\mathcal{D}_{n,\text{bulk}}^{\star}(w) = \det(\mathbf{I} - w(B_n^{\star})^2). \quad (43)$$

Then

$$\mathcal{D}_{n,\text{bulk}}^{\star} \longrightarrow \mathcal{D}_{\text{bulk}} \quad \text{locally uniformly on } \mathbb{C}. \quad (44)$$

For every $R \geq 0$, the explicit error is

$$\sup_{|w| \leq R} \left| \mathcal{D}_{n,\text{bulk}}^{\star}(w) - \mathcal{D}_{\text{bulk}}(w) \right| \leq R \delta_n^{\star} e^{1+R(15.418003788081082)^2 + R(15.418003788081082 + \varepsilon_n^{\star})^2}. \quad (45)$$

At $n = 65536$, the $R = 10^{-2}$ upper is 13.0265, so it is a convergence certificate rather than a useful pointwise enclosure. The same formula falls below 1 before $n = 2^{20}$ and reaches $3.682918268863141 \times 10^{-4}$ at 2^{30} . Smaller disks are already effective at the threshold: the $R = 10^{-4}$ upper is $1.131494590077742 \times 10^{-3}$.

6.1 Connection to the second regularized determinant

For $T \in \mathfrak{S}_2$, the natural one-step determinant is

$$\det_2(\mathbf{I} - zT) = \det\left((\mathbf{I} - zT)e^{zT}\right) = \prod_j (1 - z\lambda_j)e^{z\lambda_j}. \quad (46)$$

This was the regularization used in the long-cycle analysis [3]. The two-step determinant is its exact symmetric completion.

Proposition 6.2 (Symmetric $\det_2$ identity). *For every $T \in \mathfrak{S}_2(H)$ and every $z \in \mathbb{C}$,*

$$\det_2(\mathbf{I} - zT) \det_2(\mathbf{I} + zT) = \det(\mathbf{I} - z^2 T^2). \quad (47)$$

In particular,

$$\det_2(\mathbf{I} - z\mathcal{B}) \det_2(\mathbf{I} + z\mathcal{B}) = \mathcal{D}_{\text{bulk}}(z^2). \quad (48)$$

Proof. Both $(\mathbf{I} - zT)e^{zT} - \mathbf{I}$ and $(\mathbf{I} + zT)e^{-zT} - \mathbf{I}$ are trace class. All factors are functions of T and commute, so

$$((\mathbf{I} - zT)e^{zT})(\mathbf{I} + zT)e^{-zT} = \mathbf{I} - z^2 T^2.$$

Multiplicativity of the ordinary Fredholm determinant on $\mathbf{I} + \mathfrak{S}_1$ proves the identity. $\square$

The identity removes a possible regularization ambiguity. The linear counterterms $e^{\pm zT}$ cancel exactly. Equivalently, the trace series on the left keeps only the even coefficients:

$$\begin{aligned} \log \det_2(\mathbf{I} - zT) + \log \det_2(\mathbf{I} + zT) &= - \sum_{m \geq 2} \frac{z^m + (-z)^m}{m} \text{tr}(T^m) \\ &= - \sum_{j \geq 1} \frac{z^{2j}}{j} \text{tr}(T^{2j}). \end{aligned} \quad (49)$$

This is exactly (42) with $w = z^2$.

Remark 6.3 (Spectral information retained and lost). The zeros of $\mathcal{D}_{\text{bulk}}(w)$ encode the nonzero eigenvalues of $\mathcal{B}^2$, with algebraic multiplicity. The squaring map combines the λ and $-\lambda$ channels and therefore does not retain their one-step sign. This is appropriate for the physical two-step operator but must not be confused with a self-adjoint spectral realization.

7 Stored sparse determinant pilot

The theorem above does not require a numerical determinant evaluation. We nevertheless audit the three stored adaptive matrices to test whether the intrinsic determinant is numerically stable under the two archived dyadic refinements.

Let P be one stored sparse matrix, let $R, L \in \mathbb{R}^{n \times 2}$ be its archived right and left peripheral factors, and let $\Lambda = \text{diag}(\lambda_+, \lambda_-)$. The stored bulk center is

$$B = P - R\Lambda L^T. \quad (50)$$

For a real parameter s with $|s| < 1$, set $A_s = \mathbf{I} - sP$. The matrix determinant lemma gives

$$\det(\mathbf{I} - sB) = \det(A_s) \det(\mathbf{I}_2 + sL^T A_s^{-1} R\Lambda). \quad (51)$$

Thus the dense rank-two subtraction never needs to be materialized. Two sparse LU factorizations give

$$\det(\mathbf{I} - wB^2) = \det(\mathbf{I} - \sqrt{w}B) \det(\mathbf{I} + \sqrt{w}B). \quad (52)$$

The archived biorthogonality errors are below 4.30×10^{-15} , and the maximum relative left/right eigenvector residual is below 3.52×10^{-15} . These figures describe binary64 centers. The preceding papers separately certify that the stored peripheral factors correspond to the true spectral terms, but the LU determinants computed here are not interval-enclosed.

n	$\text{tr } B_n$	$\mathcal{D}_n(10^{-4})$	$\mathcal{D}_n(10^{-3})$	$\mathcal{D}_n(10^{-2})$
2048	0.3598968178381147	1.000079193893711	1.000792450343546	1.007975861190471
4096	0.3598916069810533	1.000079192866708	1.000792440045341	1.007975756517583
8192	0.3598903050128218	1.000079192611279	1.000792437474647	1.007975730366091

Table 2: Floating sparse-LU pilot for the stored fixed-eight-sigma matrices. At these three dimensions the fixed family coincides with the adaptive schedule. Values are diagnostics, not rigorous enclosures.

The consecutive absolute differences at $w = 10^{-2}$ are

$$1.046728883569159 \times 10^{-7}, \quad 2.615149163887054 \times 10^{-8}, \quad (53)$$

and their ratio is 0.24984. Across the five values $w = 10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}$, the mean ratio is 0.2493. This is consistent with a second-order midpoint center, as expected from the smooth Nyström analysis. It does not improve the rigorous $O(n^{-1})$ theorem, whose first-order term comes from comparing the piecewise-constant lift with the entire continuum operator in Hilbert–Schmidt norm.

The finite-dimensional symmetric $\det_2$ identity was also evaluated at every level and every pilot radius. The maximum binary64 discrepancy was $2.220446049250313 \times 10^{-16}$.

8 Certificate architecture and reproducibility

The certificate is a composition of already validated contour and kernel objects with the new trace-ideal ledger. Its main gates are:

1. the RH-42 Hilbert–Schmidt envelope for k and its derivatives;
2. the RH-43 continuum parity resolvent and weighted Schur transport;
3. the RH-44 Perron contour and intrinsic rank-two kernel bound;
4. continuum-to-Galerkin Hilbert–Schmidt Poincaré control;
5. midpoint, exact-normalization, and adaptive-cutoff Frobenius control;
6. rank-at-most-two conversion for each weighted-term difference;
7. Hilbert–Schmidt-to-trace-norm squaring;
8. even-trace and determinant continuity bounds.

The displayed ledger contains dimensions $2^{16}, \dots, 2^{30}$. These rows are convenient checkpoints, not a restriction to dyadic grids. The analytic defect formulas are valid for every integer $n \geq 65536$, and their monotonicity closes the all-grid theorem.

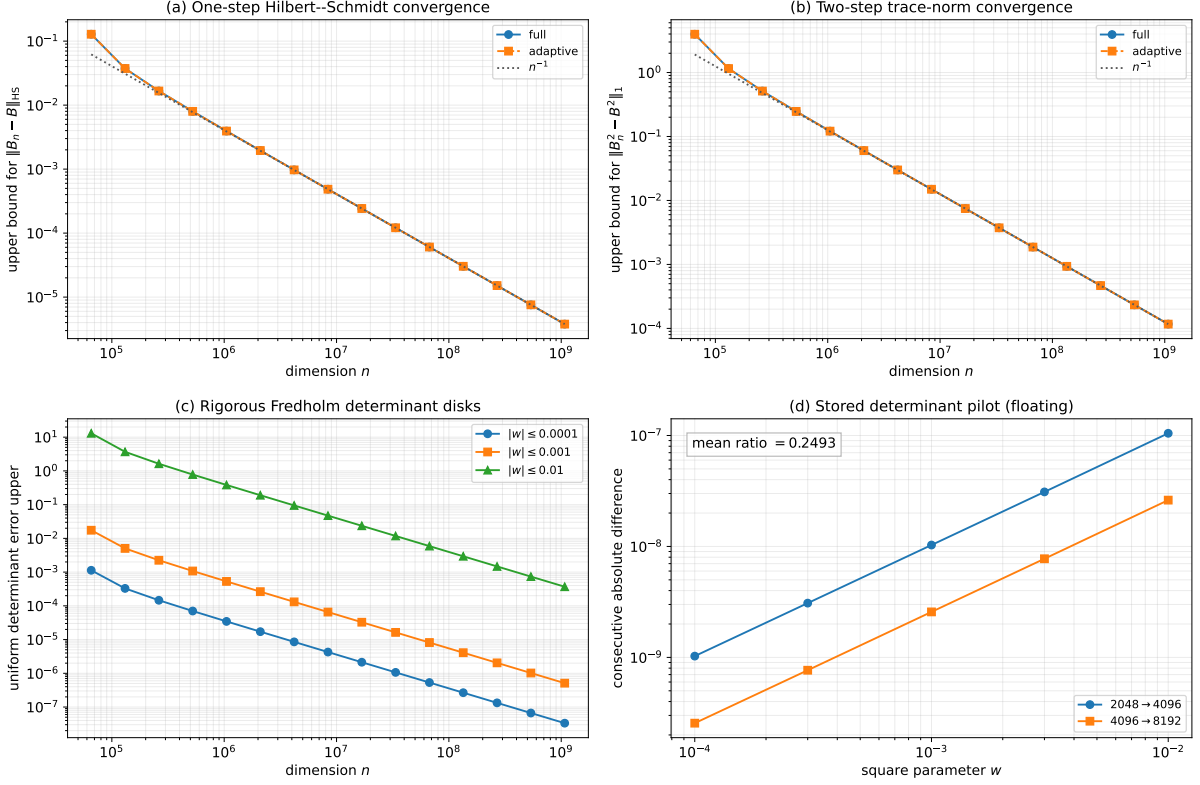


Figure 1: Trace-ideal completion of the intrinsic bulk. (a) Outward Hilbert–Schmidt errors for the full and adaptive one-step bulk families. (b) The resulting trace-norm errors for their squares. Both carry the first-order cellwise Galerkin rate. (c) Rigorous local-uniform Fredholm determinant bounds on three w -disks. (d) Floating consecutive differences of the stored sparse determinants; their approximately quarter scaling is diagnostic and is not used in panels (a)–(c).

The archive hashes every consumed external certificate, local source, result, figure, manuscript, and publication PDF. The complete replay is listed in the repository README. Rebuilding the rigorous trace ledger, tests, figures, and archive takes seconds. Rebuilding the stored 8192 determinant pilot takes approximately four minutes on the reference server because each of the five radii requires two sparse LU factorizations.

9 What is closed and what remains

The present result closes a precise functional-analytic gap.

1. The intrinsic one-step bulk now converges in the natural Hilbert–Schmidt topology for full and adaptive discretizations.
2. Its square converges in trace norm, with a completely explicit first-order bound.
3. Every fixed even bulk trace has a rigorous continuum limit.
4. The intrinsic two-step Fredholm determinant is the local-uniform limit of computable finite determinants.
5. The determinant is exactly the symmetric product of the two regularized one-step determinants from the long-cycle framework.

Several harder bridges remain open.

1. The theorem fixes $\sigma = 10^{-2}$. Uniform trace-ideal control as $\sigma \downarrow 0$ is not proved.
2. The fixed eight-sigma family is spectrally stable but is not a continuum-convergent family in the norm used here.
3. No arithmetic formula identifies $\text{tr}(\mathcal{B}^{2^m})$ with a prime or prime-power coefficient.
4. No $T \log T$ zero-counting law has been derived for $\mathcal{D}_{\text{bulk}}$ under any scaling.
5. The nonnormal operator $\mathcal{B}$ is not a self-adjoint Hilbert–Pólya operator. Squaring does not change that fact.

A natural next step is a two-parameter audit: retain the rigorous trace-ideal topology while allowing the noise width and grid dimension to vary jointly. Such a result would determine whether the intrinsic two-step determinant has a controlled small-noise limit or whether a new renormalization is required. Either outcome is mathematically informative.

The current theorem should therefore be read as infrastructure, not as an arithmetic identification. It constructs a strict intrinsic cycle-trace and spectral-determinant object at fixed noise and proves that the archived finite-dimensional route converges to it in the correct ideal topology.

Data and code availability

All source code, outward certificates, stored binary64 inputs, figures, tests, dependency hashes, archive verification, and the manuscript source are available in the accompanying repository directory https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-45-bulk-two-step-trace-norm-determinant.

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